

Aspect-Aware Sentiment Mining and LLM-Driven Ad Generation on Amazon Musical Instrument Reviews

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Abstract

We present a two-stage NLP pipeline applied to Amazon Musical Instrument reviews. *Stage 1* combines LDA topic modeling with six sentiment analysis approaches (across eight models) to extract feature-level signals: which product attributes customers praise or criticize. *Stage 2* uses these signals to generate targeted advertisements with a large language model, contrasting a *strength-amplifying* strategy (highlighting praised features) against a *problem-solving* strategy (addressing criticisms); an independent LLM judge evaluates the ads via pairwise and pointwise protocols. On the guitar corpus (134,068 reviews), fine-tuned RoBERTa (10) achieves the highest document-level Macro F1 (0.944), while BART zero-shot NLI (8) leads among training-free methods (0.911). Topic-level analysis identifies shipping damage as the most criticized feature (82.6% negative) and acoustic tone the most praised (96.2% positive). In ad evaluation, the problem-solving strategy wins 6/9 pairwise matchups and leads on all four pointwise dimensions, consistent with Prospect Theory (6). We replicate the full pipeline on *drums* and *keyboards*: Strategy B again wins both subcategories (5/8 pairwise wins, leading on every pointwise dimension), confirming that the problem-solving advantage generalizes beyond guitars.

1 Introduction

Customer reviews on Amazon contain detailed, product-specific opinions, but most marketing pipelines use them only as star-rating averages. We ask two linked questions: (1) *Which features of musical instruments do customers consistently praise or criticize?* and (2) *Can feature-level sentiment drive more persuasive advertising copy?*

Our pipeline operates on the Amazon Reviews 2023 Musical Instruments dataset (1). We first apply Latent Dirichlet Allocation (LDA) (2) to discover product feature topics, then benchmark eight sentiment classifiers to validate reliable sentiment

detection in this domain. We then perform topic-level sentiment analysis to rank features by customer satisfaction. In Stage 2 we generate pairs of ads per product, one amplifying praised features (*Strategy A*) and one addressing criticisms (*Strategy B*), and evaluate them with an independent LLM judge. The guitar subcategory serves as our primary corpus; we then replicate the entire pipeline on drums and keyboards (Section 6) to test cross-instrument generalisation.

2 Dataset and EDA

Dataset. We use Amazon Reviews 2023 Musical Instruments (1), joining reviews with product metadata on `parent_asin` (100% join rate). The full dataset contains 3,017,439 reviews across 14 subcategories with $\geq 5,000$ reviews. We select the **Guitars** subcategory (134,068 reviews) based on a multi-criteria scorecard balancing corpus size, review length, and sentiment balance, where guitars score second overall (5.74/10) with a favorable size and topic diversity.

EDA key findings. The rating distribution is heavily skewed (about $6.1\times$ more positive than negative reviews), so we use Macro F1 as the primary metric since accuracy alone is misleading. Most reviews (92.1%) are verified, and verified vs. non-verified samples show no systematic difference large enough to warrant removal. Reviews are short (median 23 words), and roughly a quarter contain fewer than 10 words, motivating our preprocessing filter.

Binary labels and split. Ratings 4-5 map to *positive*, ratings 1-2 to *negative*, and rating 3 is dropped as ambiguous. We use a stratified 80/20 split (test set: 24,815 reviews). Full EDA tables and the rating-3-free split footnote are in Appendix F.

3 Topic Modeling

We explored BERTopic (12) (all-mpnet-base-v2 + UMAP + HDBSCAN) but obtained $\approx 50\%$ outlier reviews regardless of parameter settings, consistent with known limitations of density clustering on short, mixed-register text. We therefore proceeded with LDA.

Choice of K . We grid-searched the number of topics over $K \in \{5, 10, \dots, 25\}$ using C_v coherence (11). Coherence increases monotonically with K in our range and peaks at $K=25$ (coherence = 0.541), which we adopt as our base model. The absolute coherence ceiling near 0.54 is a known property of short Amazon review text rather than a modeling failure. Other LDA hyperparameters (α, η , passes) are tuned at the same time but reported in Appendix F.

Topic labeling and assignment. From the 25 raw topics we manually retain the 13 interpretable ones and merge two accessory-related topics, giving **14 final topics** (e.g., *Tuning stability*, *String quality*, *Shipping damage*, *Beginner learning*). Each review is then assigned to its highest-probability topic within the 14 above a small probability threshold, which yields **100,248 assigned reviews (74.8%)**.

4 Sentiment Analysis

4.1 Document-Level Classification

We benchmark six approaches (eight models) on the same test set, using **Macro F1** as the primary metric.

Classical baselines. TF-IDF + Logistic Regression (LR) with bigrams achieves Macro F1 = 0.893 (tuning details in Appendix F). Sentence Transformer embeddings (all-mpnet-base-v2) (9) with LR improve marginally to 0.898.

Pre-trained models (zero-shot). nlptown-BERT, binarized by summing $P(4\star) + P(5\star)$, achieves 0.901. CardiffNLP-RoBERTa underperforms (0.875), likely due to Twitter domain mismatch. BART-MNLI (8) with NLI hypothesis “*The sentiment of this review is {positive / negative}*” achieves **0.911**, best among training-free methods. DeBERTa-NLI reaches 0.901.

Fine-tuned and few-shot. Fine-tuned roberta-base (10) achieves **Macro F1 = 0.944**, the highest overall. Mistral-7B-Instruct-v0.3

Table 1: Document-level sentiment on the guitar test set. Macro F1 is the primary metric; Neg F1 reflects minority-class performance.

Method	Acc	Mac F1	Neg F1
TF-IDF + LR	0.948	0.893	0.816
SentTransf. + LR	0.950	0.898	0.825
nlptown BERT	0.944	0.901	0.835
CardiffNLP RoBERTa	0.933	0.875	0.790
BART zero-shot	0.952	0.911	0.851
DeBERTa zero-shot	0.950	0.901	0.831
Fine-tuned RoBERTa	0.972	0.944	0.905
Mistral-7B 5-shot	0.959	0.924	0.874

(3) with 5-shot in-context learning (15) reaches 0.924, only 0.020 below fine-tuned RoBERTa despite using no labeled training data. Training and prompt details are in Appendix F.

Full results are in Table 1. Methods cluster into three tiers: (i) *top* (Macro F1 ≥ 0.92): fine-tuned RoBERTa and Mistral-7B; (ii) *upper* (0.90-0.92): BART, DeBERTa, nlptown, ST+LR; (iii) *lower* (< 0.90): TF-IDF+LR and CardiffNLP. Across topics, *Tuning stability* is consistently easiest and *Pick-ups* hardest for all methods.

4.2 Error Analysis

We inspect the 701 errors (2.8%) made by fine-tuned RoBERTa on the guitar test set to understand systematic failure modes. False positives (358) and false negatives (343) are nearly balanced, indicating no directional bias. Notably, 84% of errors are *high-confidence* mistakes (predicted probability > 0.90 or < 0.10), showing that failures are not concentrated at the decision boundary but reflect systematic, recurring patterns in the data.

Per-topic error rates. Error rates vary substantially by topic, with shipping- and service-related topics hardest and acoustic-tone topics easiest (full breakdown in Appendix F). A useful diagnostic is *Pickups*: its error rate is low (1.5%) but its Macro F1 is the lowest of any topic (0.855), because only 38 of 1,477 Pickups test reviews are negative, so a handful of minority-class mistakes depresses Macro F1 without moving accuracy.

Label noise: rating-text mismatch. Many errors come from reviews whose text contradicts the star rating, e.g. a 4-star review whose text says “*strings come out of tune so easily, also makes a weird noise on the amp?*”. The model correctly infers negative sentiment but is penalised by the positive label, exposing a fundamental limit of rating-

derived labels for fine-grained sentiment.

Mixed sentiment. The second pattern involves reviews containing genuine positive and negative signals for *different* aspects, e.g. a 5-star review reading “*Everything is amazing with this guitar!! But, the amp is absolute junk!*” where the negative half drives the model’s prediction. This is the primary motivation for topic-level sentiment analysis (Section 4.3): document-level labels cannot capture within-review variation across aspects.

4.3 Topic-Level Sentiment Analysis

Document-level labels cannot capture within-review sentiment variation, since a reviewer may praise acoustic tone while criticizing string quality. We therefore employ two approaches to topic-level sentiment.

Targeted sentiment via LLM (primary). Mistral-7B with 5-shot prompting classifies sentiment *specifically toward each review’s dominant LDA topic* for LDA-assigned reviews (excluding 414 reviews with text length ≤ 10 characters). Because LDA has already established topic relevance, no confidence threshold is needed. Results are in Table 2.

ABSA via zero-shot NLI (secondary). Using BART-large-MNLI we classify every review against every one of the 14 topic hypotheses, keeping only assignments above a confidence threshold (see Appendix F). All 14 topics score net positive under ABSA, compressed into the narrow band of 65.6%-80.6%, which we attribute to the dataset’s strong positive skew (80% positive reviews) pushing absolute scores toward the positive pole. By contrast, the LLM approach yields a far wider spread (17.4%-96.2%), with five topics scoring net negative. The two approaches agree only at the extremes (acoustic tone ranks first in both; shipping damage ranks last in both) but diverge substantially in the middle, for example Fret/neck setup (76.1% ABSA vs. 43.5% LLM) and String quality (74.6% vs. 27.6%). We read this as a methodological artifact: ABSA scores every review against every topic regardless of relevance, while the LLM approach operates only on LDA-confirmed assignments. We therefore use LLM-based targeted sentiment as the primary source for downstream analysis, and treat the ABSA agreement at the extremes as a sanity check only.

Table 2: Topic-level sentiment distribution (guitars). Source: Mistral-7B targeted sentiment on LDA-assigned reviews.

Topic	N	Pos%	Neg%
Acoustic tone	3,649	96.2	3.8
Beginner learning	12,509	91.3	8.7
Pickups	5,162	88.8	11.2
Guitar size	4,594	82.0	18.0
Playability / chords	4,437	81.4	18.6
Setup / action	3,844	79.9	20.1
Visual appearance	3,342	68.9	31.1
Accessories	7,147	67.7	32.3
Electronics / controls	1,438	62.8	37.2
Customer service / ret.	5,466	47.3	52.7
Fret / neck setup	10,203	43.5	56.5
Tuning stability	5,053	33.2	66.8
String quality	4,820	27.6	72.4
Shipping damage	4,497	17.4	82.6

5 Ad Generation and Evaluation

5.1 Ad Generation

Product and topic selection. Rather than fixing topics at market level, we take a product-first approach so that both strategies receive equally rich source material. For each product we keep only topics with at least 5 positive and 5 negative reviews, rank products by how many such topics they have, deduplicate by brand, and pick the top three; within each product we then pick the three topics closest to a 50/50 positive/negative split. The full eligibility numbers and the scoring formula are in Appendix E.

The three selected guitars are entry-level products: the Davison Full Size Electric Guitar, the Directly Cheap Acoustic Pack, and the Crescent MG38 Acoustic Starter Pack. *String quality* appears in all three, which lets us compare strategies on the same topic across different products.

Two-strategy generation. GPT-5.4 generates all ads in a two-step pipeline inspired by Chain-of-Thought prompting (4): Step 1 extracts the most compelling genuine praise (*insight_a*) and the most common real concern (*insight_b*); Step 2 generates both strategies in a single call with both insights as context, preventing strategy convergence.

Strategy A (strength-amplifying) adopts a celebratory, aspirational tone grounded in *insight_a*, mapping to Rossiter-Percy transformational motivation (14). *Strategy B* (problem-solving) uses a reassurance framing with the signature “*Not every [product] does X, but this one does*”, mapping to informational motivation. Total: 3 products $\times$ 3

Table 3: Ad evaluation results. Pairwise: win counts (9 matchups). Pointwise: mean scores (1-5 scale).

	Strat. A	Strat. B
<i>Pairwise (primary)</i>		
Wins (Ties: 2)	1 / 9	6 / 9
<i>Pointwise (supplementary)</i>		
Emot. Resonance	2.78	3.33
Argument Strength	2.56	3.00
Specificity	2.56	3.00
Purchase Motiv.	2.89	3.33
Composite	2.69	3.17

topics $\times$ 2 strategies = 18 ads.

5.2 LLM-as-Judge Evaluation

DeepSeek-V4-Flash serves as judge. Using a different company and architecture from the generator strongly mitigates self-enhancement bias (5). Strategy labels are withheld from the judge, and ad order is randomized to control for position bias.

Pairwise evaluation (primary). Each of the 9 product $\times$ topic pairs is evaluated twice (A-first and B-first) and results averaged. Strategy B wins 6/9 matchups (66.7%); 2 ties; 1 Strategy A win.

Pointwise evaluation (supplementary). Each ad is scored independently on four dimensions (1-5): Emotional Resonance (ELM peripheral route), Argument Strength (ELM central route) (13), Specificity, and Purchase Motivation (AIDA). Chain-of-thought reasoning is required before scoring.

Results. Table 3 summarises both protocols. Both methods consistently favor Strategy B (composite 3.17 vs. 2.69, $\Delta=0.48$).

Discussion. Strategy B winning on Emotional Resonance goes against ELM (13), which predicts the aspirational (transformational) strategy should have the emotional edge. A possible explanation is loss aversion (6): the “Not every X...” contrast frame triggers fear of a bad outcome, which produces a stronger emotional response than direct aspiration. The single Strategy A win (Crescent MG38 $\times$ String quality) fits the mixed gain/loss framing evidence in the literature (7): gain framing can outperform problem framing when positive aspiration is more credible than implied problem acknowledgment for a specific product-topic pair.

6 Replication: Drums and Keyboards

To assess generalisability we apply the identical pipeline to the *Drums & Percussion* and *Keyboards*

Table 4: Cross-subcategory ad evaluation summary. Pairwise: A/B/Tie counts. Pointwise: composite mean (1-5).

Subcategory	Pairwise	A	B
Guitars	1 / 6 / 2	2.69	3.17
Drums	1 / 5 / 2	2.81	3.56
Keyboards	0 / 5 / 3	2.56	3.12

& *MIDI* subcategories of the same dataset, whose feature landscapes (*drums*: percussion hardware and meditation chimes and *keyboards*: key action and built-in sounds) differ markedly from guitars.

LDA and topic-level sentiment. The same grid search yields $K=10$ for drums (coherence = 0.656) and $K=25$ for keyboards (coherence = 0.576); the lower drums optimum reflects a smaller, more thematically concentrated corpus. After manual labeling we retain **8 topics** for drums (e.g., *Drum hardware/shells*, *Product defects/returns*) and **14 topics** for keyboards (e.g., *Key feel/weighted keys*, *Returns/broken keys*). Each LDA-assigned review is scored with the fine-tuned RoBERTa classifier from Section 4.1. The most criticised topic is *Product defects/returns* for drums (55.4% negative) and *Returns/broken keys* for keyboards (68.6% negative). Functional-quality issues thus dominate the negative pole in every subcategory, mirroring guitars (full distributions in Appendix C).

Ad generation and evaluation. We re-run the product-first selection and generate ads for the top three products per subcategory (8 product $\times$ topic pairs each, as product per subcategory has only 2 topics meeting the review count threshold). Strategy B wins both replications and leads on all pointwise dimensions (Table 4), with margins larger than for guitars ($\Delta=0.75$ drums, $\Delta=0.56$ keyboards vs. $\Delta=0.48$). The lone drums Strategy A win (Woodstock Chimes $\times$ Cymbals) parallels Crescent $\times$ String quality on guitars.

7 Conclusion

Across guitars, drums, and keyboards, problem-solving ads consistently outperformed strength-amplifying ones, with functional-quality issues proving the most reliable hook for persuasion, though future work should validate these findings with larger samples and human evaluators.

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A Per-Topic Breakdown (All Methods)

Table 5 reports per-topic Macro F1 for all document-level methods. *Pickups* is consistently hardest; *Tuning stability* consistently easiest.

B ABSA vs. LLM Topic Sentiment Comparison

Table 6 compares the two topic-level sentiment approaches. ABSA scores are systematically inflated by the dataset’s positive skew; the two methods agree at the extremes but diverge in the middle.

C Replication: Per-Topic Sentiment

Tables 7 and 8 report the full topic-level positive/negative distributions for the drums and keyboards corpora, scored with the fine-tuned RoBERTa classifier on the LDA-assigned reviews. Functional-quality topics (*Product defects/returns*, *Returns/broken keys*, *Durability/stopped working*) again sit at the bottom of the ranking, mirroring the *Shipping damage / String quality / Tuning stability* cluster in guitars.

D Pointwise Rubric and Two-Step Prompt

Pointwise rubric (1–5, judge anchored). **Emotional Resonance** (ELM peripheral / Rossiter–Percy): 1 = no emotional cue, 3 = mild/generic affect, 5 = vivid and on-strategy. **Argument Strength** (ELM central): 1 = unsupported claim, 3 = mixed specific/generic, 5 = concrete causal reasoning. **Specificity**: 1 = could describe any product, 3 = balanced, 5 = product-specific terms throughout. **Purchase Motivation** (AIDA desire/action): 3 is deliberately calibrated as “neutral” to avoid optimistic drift on a 2–3 sentence ad format.

Table 5: Per-topic Macro F1, guitar test set.

Topic	TF-IDF	ST+LR	nlptown	Cardiff	BART	DeBERTa	FT-Rob.	Mistral
Accessories	0.803	0.838	0.862	0.838	0.848	0.853	0.914	0.885
Acoustic tone	0.821	0.872	0.808	0.785	0.864	0.810	0.876	0.927
Beginner learning	0.861	0.887	0.902	0.859	0.887	0.897	0.937	0.908
Customer service / ret.	0.868	0.873	0.876	0.848	0.913	0.894	0.930	0.917
Electronics / controls	0.816	0.837	0.879	0.794	0.820	0.821	0.908	0.868
Fret / neck setup	0.884	0.880	0.887	0.844	0.908	0.892	0.939	0.906
Guitar size	0.760	0.825	0.886	0.790	0.836	0.814	0.886	0.896
Pickups	0.824	0.763	0.711	0.692	0.776	0.744	0.855	0.773
Playability / chords	0.855	0.859	0.856	0.816	0.882	0.865	0.942	0.893
Setup / action	0.809	0.830	0.805	0.817	0.838	0.839	0.912	0.903
Shipping damage	0.887	0.874	0.899	0.889	0.913	0.888	0.929	0.912
String quality	0.894	0.898	0.891	0.869	0.906	0.905	0.948	0.907
Tuning stability	0.924	0.935	0.950	0.921	0.950	0.933	0.974	0.944
Visual appearance	0.845	0.847	0.845	0.801	0.867	0.838	0.913	0.872

Table 6: Topic positive% under ABSA (BART zero-shot) vs. LLM few-shot (Mistral-7B). Sorted by LLM positive%.

Topic	ABSA Pos%	LLM Pos%
Acoustic tone	80.6	96.2
Beginner learning	79.2	91.3
Pickups	76.9	88.8
Guitar size	79.2	82.0
Playability / chords	79.6	81.4
Setup / action	78.4	79.9
Visual appearance	78.5	68.9
Accessories	70.6	67.7
Electronics / controls	77.4	62.8
Customer service / ret.	70.7	47.3
Fret / neck setup	76.1	43.5
Tuning stability	73.2	33.2
String quality	74.6	27.6
Shipping damage	65.6	17.4

Two-step ad generation (per product×topic).

Step 1 (analysis). The generator reads up to 10 positive and 10 negative real reviews for the topic and outputs insight_a (most compelling genuine praise) and insight_b (most common real concern with a feasible counter). *Step 2 (generation).* A single call returns Strategy A and Strategy B together, with both insights as context, forcing active contrast and preventing convergence. Step 1 is the chain-of-thought scaffold and is grounded in actual review evidence rather than free generation.

E Product×Topic Selection Rule

For every subcategory we (i) enumerate all product×topic pairs with ≥ 5 positive and ≥ 5 negative reviews; (ii) rank products by score = $n_{\text{qualifying_topics}} + \text{description_length} / 1000$; (iii) deduplicate by brand keeping the highest-scoring product; (iv) retain the top three

Table 7: Topic-level sentiment, drums.

Topic	N	Pos%	Neg%
Gift / holiday purchase	23,389	97.0	3.0
Beginner / ease of play	12,469	92.1	7.9
Singing bowls / meditation	15,278	91.3	8.7
Drum hardware / shells	10,335	88.1	11.9
Cymbals	10,587	87.6	12.4
Small percussion accessories	15,261	85.5	14.5
Electronic drums / practice pad	9,753	82.2	17.8
Product defects / returns	16,646	44.6	55.4

Table 8: Topic-level sentiment, keyboards.

Topic	N	Pos%	Neg%
Gift / holiday purchase	6,100	99.1	0.9
Kids music lessons	2,559	97.1	2.9
Synthesizer / sound design	2,267	92.3	7.7
Beginner learning	7,742	90.8	9.2
MIDI controller / live perform.	1,577	89.7	10.3
Key feel / weighted keys	4,979	87.8	12.2
Brand quality / acoustic feel	4,814	83.8	16.2
Sustain pedal	826	82.4	17.6
Sound quality / speaker	3,746	80.1	19.9
Appearance / color / display	1,293	78.3	21.7
Connectivity (USB/MIDI/phone)	1,676	68.4	31.6
Power supply & accessories	3,581	66.8	33.2
Durability / stopped working	1,741	57.3	42.7
Returns / broken keys	5,696	31.4	68.6

products with at least 2 qualifying topics and a non-empty metadata description; (v) for each retained product pick the (up to) 3 qualifying topics minimising $|\text{pct_positive} - 50\%|$ to maximise input parity between strategies. This is fully automated (no manual curation), so the procedure ports unchanged across guitars, drums, and keyboards.

F Reproducibility and Resources

Splits. All sentiment classifiers use stratified 80/20 splits with random_state=42. The default

test set holds 24,815 reviews (3,698 negative / 21,117 positive). Fine-tuned RoBERTa and Mistral-7B use the same split applied to the 24,849 rating-3-free rows, which differs by 34 samples ($<0.15\%$) and does not affect conclusions.

LDA hyperparameters. LDA is trained with Gensim using α =asymmetric, η =auto, 10 passes, evaluated with the C_v coherence metric (11). The grid search covers $K \in \{5, 10, 15, 20, 25\}$ and four (α, η) configurations. Topic assignment uses a probability threshold of 0.15.

Classical baseline tuning. TF-IDF + LR best parameters: $C=10$, $\text{max_features}=100\text{k}$, $\text{ngram_range}=(1,2)$, $\text{max_df}=0.85$. Sentence Transformer embeddings use the all-mpnet-base-v2 model.

Fine-tuned RoBERTa. RoBERTa-base, 3 epochs, learning rate 2×10^{-5} , linear warmup, early stopping on validation Macro F1.

Mistral 5-shot. Mistral-7B-Instruct-v0.3 with temperature 0 and a fixed 5-shot exemplar set drawn from the training split.

ABSA threshold. BART-large-MNLI assigns a topic label only when entailment probability is at least 0.70. Without this threshold, low-confidence predictions inflate noise.

Per-topic error rates (RoBERTa). Hardest topics: Shipping damage 7.1%, Customer service / returns 6.8%. Easiest topics: Acoustic tone 1.0%, Setup / action 1.3%.

Stage-2 API setup. Generator and judge models in Stage 2 are accessed through their public APIs at the time of writing. Ad order is randomised and each pair is judged twice (A-first and B-first) to control for position bias (5).